MineGuard

Enhancing Miner Safety with Wearable Technology

X Team

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Understanding Miner Safety

Miner safety is a critical concern in hazardous environments like mines.

What problems could the miner have?

Gas Toxicity
Low Vital Signs
Getting Lost

Proposed Soultion

A chip is embedded in the miner's clothing

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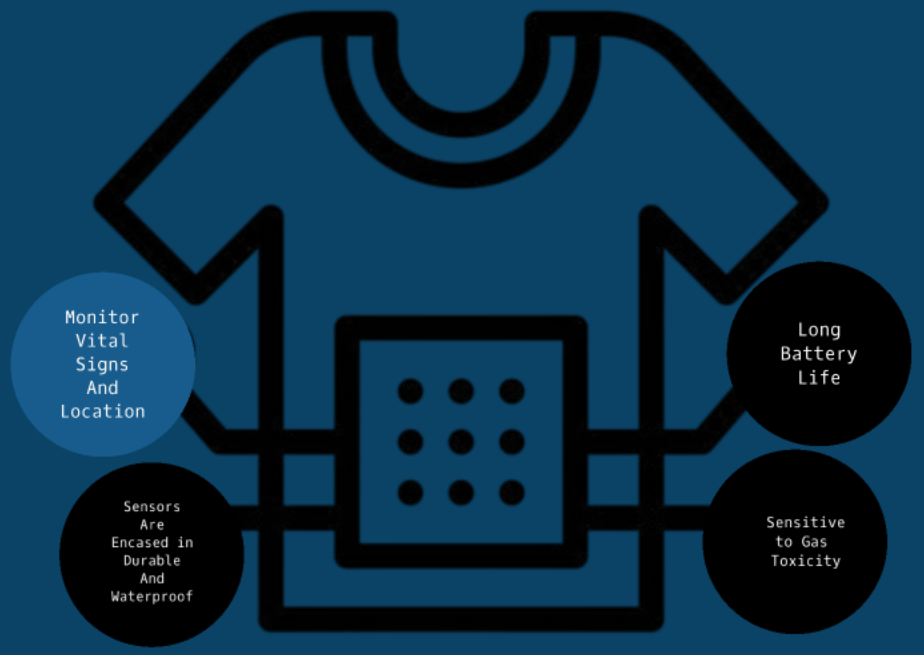
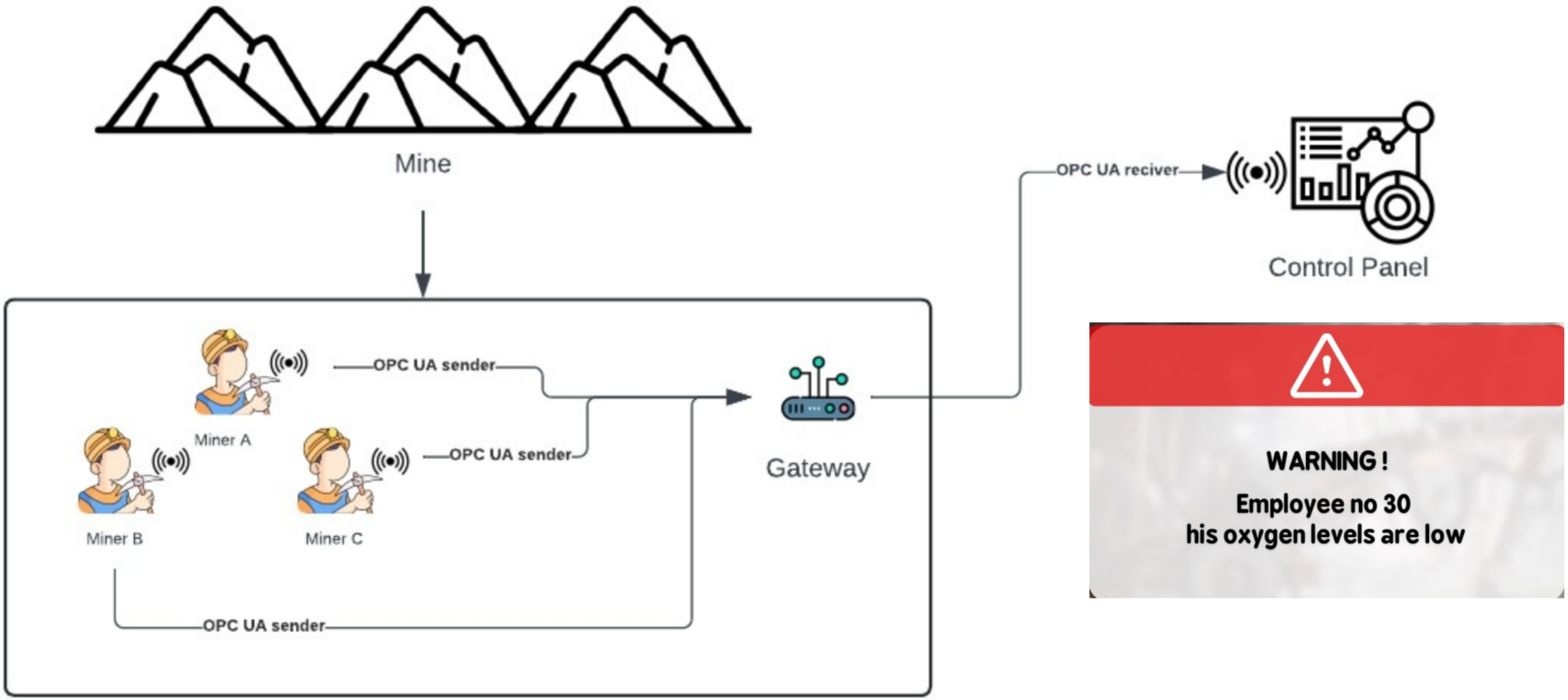


Monitor Vital Signs And Location

Sensors
Are
Encased in
Durable
And
Waterproof

Long Battery Life

Sensitive to Gas Toxicity


Mine
OPC UA sender
Miner A
OPC UA sender
Miner B
Miner C
OPC UA sender
Gateway
OPC UA reciver
Control Panel
WARNING !
Employee no 30
his oxygen levels are low

Website

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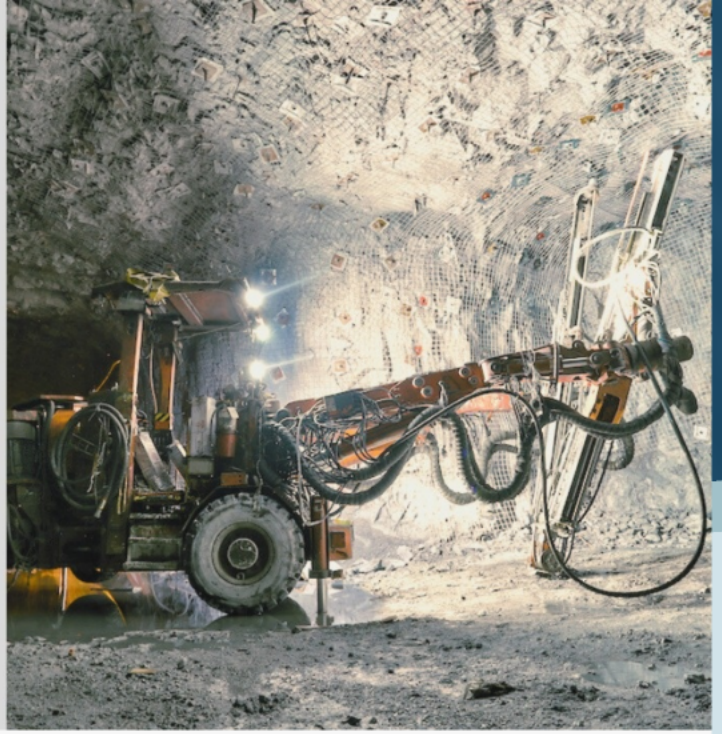
Here we provide to you what you need to get the best protection for your minors

LET'S START



Pricing Plan

Basic

Good for individual

customizable :

- T-shirt or bodysuit or bracelet

choose this plan

Premium

Good for company

Full package:

- clothing
- control panel
- provide wireless gateway
- provide safety team

choose this plan

On Demand

Ex : sea mines ...

choose this plan

Used Technology

Monitoring System

GPS/Location Tracking

Embedded Systems

Sensors

Machine Learning/ AI

Battery and Power Management

Data Processing and Storage

Wireless Communication

Internet Of Things

Business Model

Key Partners

- Mining companies (mining workers between 18 and 40 years old).
- Health sector (health centers - hospitals).

Key Activities

- Design and development of suits
- Manufacture of high quality products.
- Marketing and sales.
- Technical support and support services.
- Continuous research and development.
- Database management.

Key Resources

- Sensors.
- Wireless networks.
- Uniforms (suit - shirt - watch).

Added Value

- Care and safety for workers.
- Reduce costs of accidents and injuries.
- Increase profitability and revenues in the long term.
- Improve job satisfaction among workers.

Channels

- Presence in the scope of work of mining companies.
- Shops.
- Advertising.

Customer Relationships

- Providing a high quality product.
- Listening to customers and responding to their requirements and needs.
- Continuously developing the product to keep up with the latest technologies.
- Offering special offers to loyal customers

Customer Segments

وزارة الصناعة والثروة المعدنية
Ministry of Industry and Mineral Resources

code
مركز ريادة الأعمال الرقمية
Center of Digital Entrepreneurship

SIEMENS

Cost Structure

Costs of a single device

- Materials (sensors, electronic components, etc.): 1,550 SAR.
- Manufacturing costs: 350 SAR.
- R&D costs: 1,000 SAR.
- Administration costs: 350 SAR.

Internal wireless network costs

- 3,750 - 7,500 SAR.

Revenue Streams

- Sponsors.
- Facilities.
- Loans.
- Hackathons.

Future Work

- Advanced Data Analytics and Predictive Modeling
- Combined Health and Environmental Sensors
- Automated Emergency Response
- Ergonomic Design

Thank You

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